

# Automatic DC-Offset Calibration for a Low-Cost Geophone Front End

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**Abstract**—This paper presents a foreground calibration method for a low-cost, cascaded geophone analog front end, where amplifier and reference offsets compound stage-to-stage and can drive intermediate nodes into saturation. The proposed method sequentially observes and independently trims four cascaded stages – a programmable-gain amplifier (PGA), a band-pass filter (BP), a summing stage (SUM), and a low-pass filter (LP) – reusing existing on-chip conversion and filtering resources rather than adding dedicated calibration hardware. Oscilloscope measurements on two independently assembled prototype boards, across the PGA's full configurable gain range ( $\times 1$  to  $\times 50$ ), show that after calibration the stage nearest the ADC is left within 4% of the ADC's full-scale range in every board/gain combination tested, down from as much as 46% uncalibrated. The PGA stage converges only partially under some conditions – shown to result from the loop's deadband gating convergence at that stage specifically, not from a hardware limit – but this does not affect the headroom result, which is measured after the full cascade and dominated by the downstream stages. A full four-stage run completes in approximately 10 s typical (57.6 s worst case). The calibrated front end has additionally produced coherent multichannel surface-wave field records, demonstrating operation beyond the bench.

**Index Terms**—Geophone, analog front end, DC-offset calibration, mixed-signal systems, PSoC

## I. INTRODUCTION

Multichannel surface-wave methods use spatially distributed vibration measurements to estimate the dispersive behavior of Rayleigh waves and infer near-surface mechanical properties [1], [2]. Although the processing methods are well established, the acquisition hardware remains a significant part of the cost and deployment complexity, so compact wireless nodes built from inexpensive geophones and mixed-signal microcontrollers are attractive for dense, reconfigurable arrays [3]. Recent embedded geophone interfaces cascade similar amplifier/filter/gain stages without addressing automatic per-stage offset recovery [4].

The present work forms part of a low-cost, dynamically reconfigurable Internet-of-Things acquisition platform [5] based on SM-24 moving-coil geophones [6]. A PSoC 5LP was selected as the analog front end because it integrates programmable-gain amplifiers, operational amplifiers, analog filters, voltage DACs, analog multiplexers, a delta-sigma ADC, a digital filter block, DMA, and an Arm Cortex-M3

processor in one device [7]. The guiding principle is to reuse this already-available hardware rather than add dedicated calibration components: the same ADC and digital filter are time-shared between acquisition and calibration, avoiding a separate converter or processor, in the spirit of digitally assisted analog design [8]. This reconfigurability also imposes a limitation: its internal analog blocks cannot be independently optimized for low offset, drift, and flicker noise, the same tradeoff reported for a single-chip PSoC potentiostat [9].

The geophone conditioning circuit is derived from the cascade correction and subtraction topology of Ma *et al.* [10]; mapped to the PSoC, accumulated DC errors prevented sufficiently aggressive gain settings and became particularly severe at the SUM output.

A continuously active DC servo, evaluated as an initial solution, proved impractical (Section V); the proposed method instead performs a finite foreground calibration before measurement and disconnects the controller afterward.

Self-calibration using on-chip resources is not itself new. The PSoC-Stat injects five known current levels with an IDAC and uses the resulting ADC values to estimate the gain and offset of a TIA/ADC signal chain [9]. Infineon's AN68403 applies an internal VDAC to a PGA/ADC chain, measures the system offset with grounded inputs, and stores gain and offset corrections in EEPROM [11]. These methods primarily correct the numerical transfer characteristic at the ADC output. Digitally governed physical trimming has also been shown at the single-amplifier level (body-bias offset calibration in memristive-array readout [12]); the proposed loop instead changes the physical reference of each stage across a whole cascade, toward its intended operating point within the available DAC range and resolution.

The front end is deliberately DC-coupled rather than AC-coupled: the SM-24's usable band extends down toward its mechanical corner, and a coupling capacitor placed ahead of it would need a corner well below 1 Hz to avoid attenuating exactly the low-frequency content the acquisition targets, degrading the signal it is meant to preserve. That constraint rules out the simplest fix for offset – block it with a capacitor – and leaves per-stage trimming as the remaining option once the amplifiers themselves cannot be swapped for lower-offset

external parts without abandoning the single-chip premise, unlike the classical autozero/chopper toolbox for offset and 1/f noise [13].

A SUM-only fixed DC injection was tried first, as an early workaround (Section II): it recenters one node but cannot recover stages before or after it.

The contribution is threefold: 1) a per-stage analog calibration architecture that reuses the PSoC analog multiplexer, ADC, digital filter, and four VDACS, so that improving an existing budget-constrained analog front end does not require oversizing each stage with dedicated precision components; 2) a measurement-gated foreground implementation, using dead-band, anti-windup, slew limiting, stability detection, and one-LSB refinement, that freezes all corrections before acquisition, adding no closed-loop poles to the acquisition transfer function; and 3) an experimental validation on two independently assembled boards across the PGA's full configurable gain range ( $\times 1$  to  $\times 50$ ), showing that the calibrated front end keeps every stage within a small fraction of the ADC's full-scale range regardless of board or gain, even where an individual stage such as the PGA converges only partially. The design goal is not to null every stage's offset exactly, but to guarantee that no board, at no gain setting, requires expert per-unit tuning to stay within its acceptance range.

## II. PLATFORM DESIGN

The platform was selected for low cost, reconfigurable analog routing, hardware-assisted signal processing, and reuse of acquisition resources across operating modes. In acquisition mode the PSoC conditions and digitizes the geophone output; in calibration mode the measurement path, digital-filter coefficients, and DAC-control firmware reconfigure to close a temporary mixed-signal loop, while the input chain stays unchanged.

The front end (Fig. 1) is a measurement chain in cascade: the geophone signal is conditioned by PGA, BP, SUM, and LP in series, so an offset introduced at any stage propagates, scaled by every downstream gain, to the ADC input. Correcting this chain is therefore not a single adjustment but a sequential, per-stage process: each stage is measured and trimmed in signal-flow order before the next is addressed, so that a correction made upstream is never undone by a later stage (Section III).

An analog multiplexer connects one stage output at a time to the same 18-bit delta-sigma ADC used to digitize the geophone signal, so the acquisition converter itself serves as the error sensor without a second ADC; software-enforced mutual exclusion prevents two analog outputs from being connected at once. A switched capacitor behind the selected multiplexer MOSFET, whose on-resistance forms a limited RC anti-alias network, isolates the amplifier output from a direct capacitive load.

For the  $i$ th stage, its slowly varying output can be represented as

$$y_i^{\text{DC}} = A_i y_{i-1}^{\text{DC}} + b_i + K_i u_i, \quad (1)$$

where  $A_i$  is the stage gain,  $b_i$  the amplifier and reference errors,  $K_i$  the signed transfer from the trim DAC to the measured node,

and  $u_i$  the DAC code. Adjusting only the SUM reference – the first fix attempted, before this per-stage architecture – forces one output toward its target but leaves the remaining  $y_i^{\text{DC}}$  values uncontrolled: an earlier rail cannot be recovered, offsets appearing after SUM stay uncompensated, and the fix does not extrapolate to front ends without an explicit summing node.

The proposed architecture instead chooses each  $u_i$  in Eq. 1 independently so that every measured  $y_i^{\text{DC}}$  approaches its stage reference. In the differential geophone front end considered here, all four calibration targets correspond to zero ADC counts.

Eq. 1 is lower-triangular in stage index:  $y_i^{\text{DC}}$  depends on  $y_{i-1}^{\text{DC}}$  but not on any downstream stage. Correcting in signal-flow order (PGA, BP, SUM, LP, as in Section III) therefore guarantees  $y_{i-1}^{\text{DC}}$  is already at its target when stage  $i$  is trimmed: a single forward pass suffices, so the processing order is not arbitrary but the one that makes one-pass calibration correct.

## III. SELF-CALIBRATING GEOPHONE PLATFORM

### A. Digital Hardware and Calibration Sequence

Fig. 2 summarizes, first, the sequence run by the digital hardware described next. At boot, after a gain change, or upon command, the firmware loads the stored DAC values, verifies the offsets, and restores the measurement path without a full run if all stages are within tolerance; otherwise it processes PGA, BP, SUM, and LP sequentially, settling after each multiplexer switch and FIR-history reset, bounded by a watchdog and per-stage sample limit. During acquisition the controller is inactive, the multiplexer returns to the LP output, and all VDAC codes remain constant: the method changes the DC operating point without introducing a time-varying feedback loop into the acquired signal.

The PSoC digital filter block is reused between the two operating modes (Fig. 3): during acquisition it implements the geophone filter, and during calibration its coefficients switch to an aggressive low-pass response that provides a stable DC-level estimate for the stage being measured. Hardware reuse here reduces CPU intervention and preserves the existing acquisition architecture; acquisition coefficients and signal path are restored at the end of calibration. Successful DAC codes are stored in EEPROM, one record per PGA gain setting (hardware class, gain code, stage/valid-stage mask, four DAC codes, version, CRC); a failed run never overwrites a valid record.

### B. Control Law and Calibration Parameters

For each stage, the filtered measurement  $\hat{y}_i[n]$  produces the ADC-domain error  $e_i[n] = r_i - \hat{y}_i[n]$ , where  $r_i = 0$  for the current differential geophone configuration. The error is first mapped to the 8-bit DAC domain:

$$\varepsilon_i[n] = \text{round}\left(\frac{e_i[n] V_{\text{ADC}} N_{\text{DAC}}}{N_{\text{ADC}} V_{\text{DAC}}}\right), \quad (2)$$

with  $V_{\text{ADC}} = 5000$  mV,  $N_{\text{ADC}} = 262144$ ,  $V_{\text{DAC}} = 4080$  mV, and  $N_{\text{DAC}} = 255$  in the implemented configuration.

The calibration architecture only requires the selected stage be measurable and the trim DAC provide sufficient control

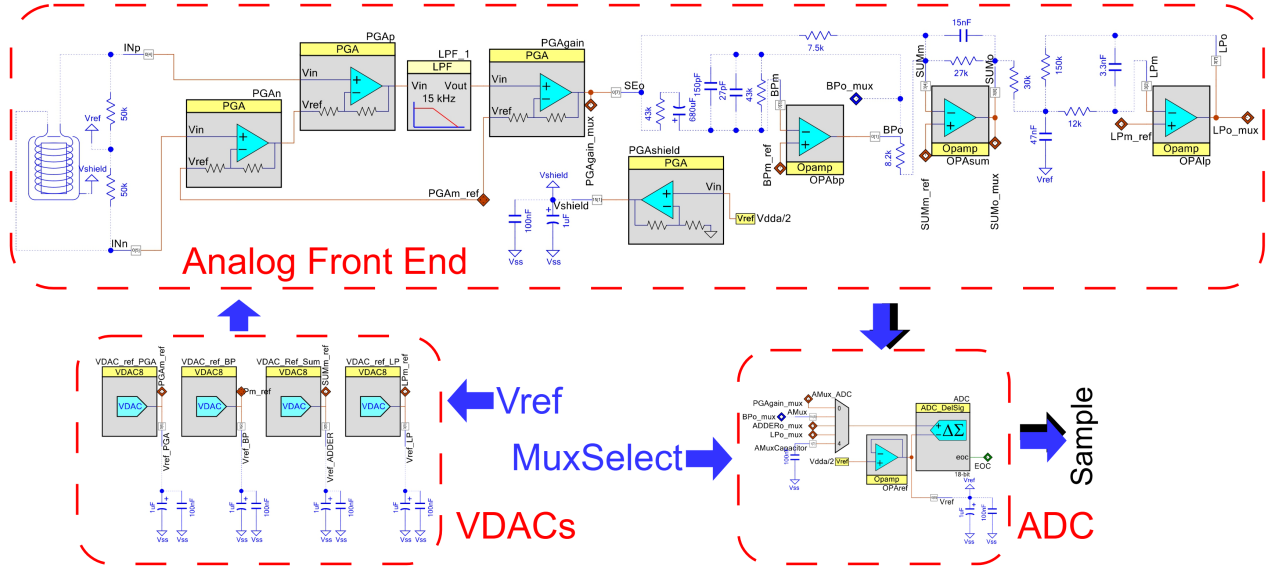


Fig. 1. Analog signal chain: four independent VDAC references feed the programmable-gain amplifier (PGA), band-pass filter (BP), summing stage (SUM), and low-pass filter (LP) cascade, with the shared ADC/multiplexer measurement path at the output.

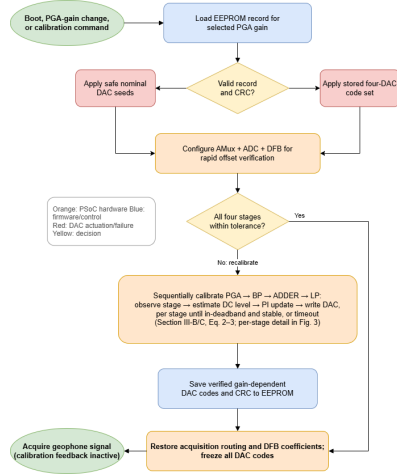


Fig. 2. Flowchart of the proposed foreground autocalibration algorithm, including stored-code verification, sequential per-stage correction, and transition to acquisition.

range, so the controller can be chosen per application. In this prototype, a positional PI law was used for its simplicity:

$$u_i^*[n] = u_{i,0} + s_i \left[ \frac{k_{p,i}}{|K_i|} \varepsilon_i[n] + \frac{k_{i,i}}{|K_i|} \sum_{q=0}^n \varepsilon_i[q] \right], \quad (3)$$

where  $u_{i,0}$  is the feedforward code, an initial DAC value near stage  $i$ 's expected operating point;  $s_i \in \{-1, +1\}$  is the injection polarity; and  $K_i > 0$  is the magnitude of the DC gain from the DAC's injection point to the shared measurement node – i.e., how much a static VDAC code change at stage  $i$  moves the measured node, expressed in ADC-equivalent counts per VDAC code (Eq. 2).  $K_i$  is a nominal design value, fixed for BP, SUM, and LP (2, 7.89, and 6, reflecting downstream gain, not individually measured per board); for PGA,  $K_i$  and  $s_i$  are recomputed from the active gain code; its feedforward

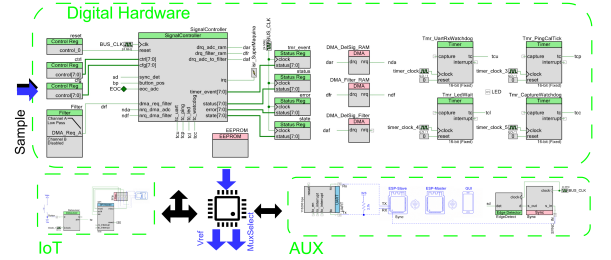


Fig. 3. Digital hardware that drives calibration: the signal controller sequences DMA transfers between the ADC, the reused digital filter block, and EEPROM, and reconfigures the shared analog multiplexer between acquisition and calibration. Black arrows mark the acquisition-time signal route; blue arrows mark the calibration-time route.

code  $u_{i,0}$  was hand-set to the DAC value for gain  $\times 50$  – the circuit's power-up gain – and left fixed. For this deployment  $k_{p,i} = 0$  for all four stages (Table I lists only  $k_{i,i}$ ): response speed is not a priority for a calibration that triggers once after deployment, so the general law reduces to a pure integrator.

The command  $u_i^*[n]$  of Eq. 3 is saturated to the valid VDAC interval and limited to one code per sample, with conditional integration for clamping anti-windup. The deadband is  $\Delta_i = \max(1, \lceil 1.2|K_i| \rceil)$  DAC codes; convergence requires the error to stay in the same quantized bucket for a configurable number of samples, and a final adjacent-code trial is kept only if it reduces the filtered error. If a stage misses its deadband before a per-stage timeout, the DAC stays at the best code found, the stage is flagged not converged, and the sequence moves on, bounding worst-case time without leaving any stage arbitrary or saturated.

Table I lists the parameters used, to support reproduction. Settle is 512 samples (0.20 s) for PGA/BP/SUM and 1024 (0.39 s) for LP; lock requires 512 or 1024 consecutive in-deadband samples, and refinement settles a further 1024–2048

TABLE I  
IMPLEMENTED PER-STAGE CALIBRATION PARAMETERS. THE DC ESTIMATOR RUNS AT THE ADC NATIVE SAMPLE RATE OF 2604 HZ.

| Stage | $K_i$          | $k_{i,i}$          | Timeout (s) | Stage | $K_i$ | $k_{i,i}$          | Timeout (s) |
|-------|----------------|--------------------|-------------|-------|-------|--------------------|-------------|
| PGA   | gain-dependent | $3 \times 10^{-4}$ | 11.5        | SUM   | 7.89  | $5 \times 10^{-4}$ | 11.5        |
| BP    | 2              | $5 \times 10^{-4}$ | 17.3        | LP    | 6     | $5 \times 10^{-4}$ | 17.3        |

samples. Timeouts are worst-case ceilings, not the typical duration shown in Fig. 4.

The 8-bit VDAC resolution suffices here; a finer step could use a PSoC IDAC/resistor or a buffered topology [14] – an implementation choice, not a limitation, since controller, range, and step size can all adapt while keeping the same measure–estimate–correct sequence.

#### IV. EXPERIMENTAL VALIDATION

##### A. Prototype and Comparison Conditions

The validation compares two configurations on two independently assembled SM-24 geophone-conditioning prototypes (Board A, Board B), each containing the four independently controlled VDAC references: *Default*, fixed nominal VDAC references without calibration, and *Proposed*, sequential four-stage calibration followed by frozen-reference acquisition. The *Default* references were hand-set once for gain  $\times 50$ , the PSoC’s power-up gain, and then left fixed across the whole gain range tested. Measurements span the PGA’s full gain range,  $\times 1$  to  $\times 50$ , rather than one representative setting, testing board- and gain-independence directly instead of assuming it from a single point.

##### B. Design Target: Headroom, Not Zero

The calibration loop is not designed to null every stage’s offset exactly; *headroom* here is the ADC full-scale margin remaining before a stage saturates, so a lower headroom-consumed figure is better. Its deadband, timeout, and controller gains are firmware parameters, not physical constants (Section III): a tighter deadband is harder to reach and easier to leave, a longer FIR estimation window improves the DC estimate at the cost of convergence time, and a longer per-stage timeout lets a conservative controller finish without needing to be aggressive. The design target that these parameters serve is that no stage, on any board, at any gain, leaves the signal at risk of railing the ADC – not that every stage converges to zero with equal precision. Section IV-D shows a case where this target is met even though one individual stage converges only partially.

##### C. Measured Headroom Recovery Across the Gain Range

For each board and gain, the DC offset at the output of the PGA, BP, SUM, and LP stages was measured with a Tektronix TDS 2004B oscilloscope (4 channel, 60 MHz, 1 GS/s), reading the mean voltage over a 100 ms window after the transient settled, both with the default (uncalibrated) VDAC references and after the proposed calibration converged. Table II reports the headroom consumed at all four stages – PGA, BP, SUM, and LP – as a percentage of the ADC’s 5 V full-scale span

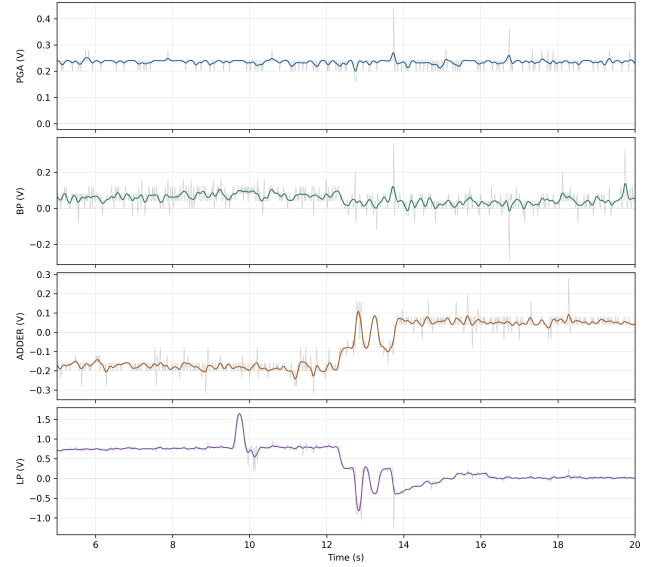


Fig. 4. Oscilloscope capture of the PGA, BP, SUM, and LP stage outputs during a single calibration run at PGA gain  $\times 16$ . Uncalibrated until  $t \approx 9$  s, all four stages converge by  $t \approx 16$  s; PGA shows a visibly larger residual, discussed in Section IV-D.

( $\pm 2.5$  V around its mid-supply reference),  $N=10$  independent runs per board/gain/stage cell. LP is the stage immediately upstream of the ADC, and therefore the one that determines whether the signal rails; PGA, BP, and SUM show where in the cascade the default offset originates and how each is reduced along the way. Board B was run at  $\times 1$  and  $\times 4$  but is excluded from those cells: at those gain/board combinations the default offset already falls inside the loop’s normal deadband, so the controller correctly reports convergence almost immediately without moving any DAC, leaving no meaningful Default-vs-Proposed contrast to report; forcing the loop to act instead required lowering the deadband by hand, which changes the test condition (Section IV-D).

Across every board/gain combination measured – all 9 board  $\times$  gain cells, LP block of Table II – calibration leaves the LP stage at  $\leq 4\%$  of the ADC’s full-scale range, down from as much as 46% uncalibrated on Board A at  $\times 1$ . Board A’s default headroom consumption falls as gain increases, while Board B starts, and stays, much closer to zero across the whole range – the boards differ in default operating point, yet calibration brings both into the same narrow band, regardless of board or starting offset. This is the paper’s central quantitative result: independent of assembly or gain setting, the proposed method consistently converts a highly variable, sometimes rail-risking default offset into a small, bounded, and predictable one.

##### D. The PGA Stage: A Partial Result, Explained

Unlike BP, SUM, and LP, the PGA stage’s post-calibration offset is not consistently much smaller than Default: calibration reduces the mean  $|\text{offset}|$  at the PGA by roughly 4–54%, smallest at  $\times 1$ . Two effects contribute. First, the deadband  $\Delta_i$  (Section III) is a firmware margin, not a physical limit: on Board B at  $\times 4$ , the default PGA offset fell inside the normal deadband on every run attempted, and the controller

TABLE II  
HEADROOM CONSUMED AT EACH STAGE – I.E., THE RESIDUAL DC OFFSET AS % OF THE ADC’S 5 V FULL-SCALE SPAN ( $\pm 2.5$  V AROUND ITS MID-SUPPLY REFERENCE), REPORTED AS DEFAULT / PROPOSED. LOWER IS BETTER.  $N=10$  PER CELL.

| Gain        | Board A                   |                           |                                                  |                                                  | Board B                   |                               |                           |                           |
|-------------|---------------------------|---------------------------|--------------------------------------------------|--------------------------------------------------|---------------------------|-------------------------------|---------------------------|---------------------------|
|             | PGA                       | BP                        | SUM                                              | LP                                               | PGA                       | BP                            | SUM                       | LP                        |
| $\times 1$  | $3.5 \pm 0.2/3.4 \pm 0.3$ | $0.8 \pm 0.2/1.2 \pm 0.2$ | <b>21.8 <math>\pm</math> 1.1</b> / $0.8 \pm 0.2$ | <b>45.8 <math>\pm</math> 3.0</b> / $3.2 \pm 0.8$ | –                         | –                             | –                         | –                         |
| $\times 4$  | $2.4 \pm 0.1/2.1 \pm 0.3$ | $4.9 \pm 0.3/1.0 \pm 0.2$ | $6.7 \pm 1.0/1.0 \pm 0.3$                        | <b>42.6 <math>\pm</math> 3.9</b> / $3.9 \pm 0.8$ | –                         | –                             | –                         | –                         |
| $\times 16$ | $2.4 \pm 0.1/1.9 \pm 0.2$ | $3.8 \pm 0.9/1.5 \pm 1.5$ | $4.9 \pm 1.5/1.4 \pm 1.5$                        | <b>29.6 <math>\pm</math> 6.5</b> / $2.3 \pm 1.3$ | $2.1 \pm 0.1/1.7 \pm 0.1$ | $0.66 \pm 0.02/0.66 \pm 0.02$ | $4.5 \pm 0.8/0.6 \pm 0.5$ | $3.0 \pm 0.1/0.7 \pm 0.1$ |
| $\times 32$ | $1.0 \pm 0.2/0.5 \pm 0.2$ | $7.0 \pm 0.4/1.0 \pm 0.5$ | $7.4 \pm 0.3/0.6 \pm 0.2$                        | <b>12.4 <math>\pm</math> 1.3</b> / $1.7 \pm 0.6$ | $1.0 \pm 0.2/0.5 \pm 0.2$ | $0.66 \pm 0.01/0.66 \pm 0.01$ | $8.0 \pm 0.8/1.2 \pm 0.3$ | $2.9 \pm 0.1/0.4 \pm 0.2$ |
| $\times 50$ | $1.2 \pm 0.3/1.0 \pm 0.5$ | $1.0 \pm 0.7/0.9 \pm 0.5$ | $0.4 \pm 0.2/0.8 \pm 0.6$                        | $4.1 \pm 1.9/1.2 \pm 0.4$                        | $2.5 \pm 0.1/1.8 \pm 0.3$ | $0.65 \pm 0.01/0.65 \pm 0.02$ | $0.9 \pm 0.6/0.9 \pm 0.6$ | $2.9 \pm 0.1/0.7 \pm 0.2$ |

All values %FS, Default / Proposed. **Red**: Default  $\geq 10\%$ FS (highest saturation risk before calibration). **Orange**: Proposed  $\geq$  Default (calibration did not reduce offset; Section IV-D). Board B  $\times 1/\times 4$  excluded: loop converged immediately, no action taken.

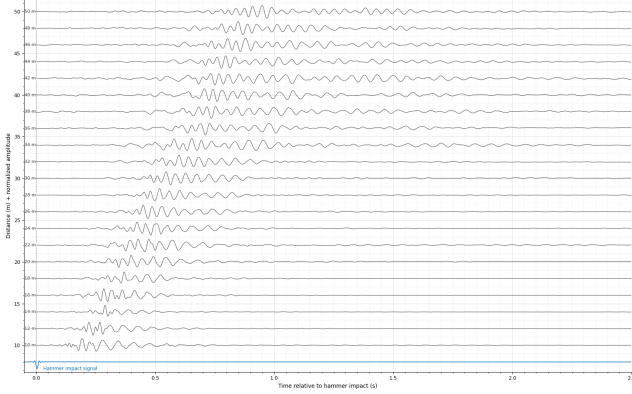


Fig. 5. Multichannel waterfall field record acquired with the calibrated front end.

declared convergence without moving the DAC – confirmed directly, not inferred – until the deadband was tightened by hand, confirming that a wide-enough deadband can gate PGA convergence specifically. Second, the FIR window used to estimate the DC level (128 samples) limits how quickly a small residual can be resolved from noise; a longer window would help, at the cost of convergence time – left for future tuning. Both point to firmware parameters, not a hardware ceiling, and neither compromises the headroom target in Table II, dominated by the LP stage, not the PGA.

Fig. 4 shows an oscilloscope capture of the four measured stage outputs during an actual run, spanning the sequential  $\text{PGA} \rightarrow \text{BP} \rightarrow \text{SUM} \rightarrow \text{LP}$  correction. Each output steps as the controller reaches it and holds once converged, visible as the successive plateaus in the SUM and LP traces; the full sequence completes in approximately 10 s.

#### E. Field Use of the Calibrated Front End

The laboratory results above are bench measurements of DC offset, not a field survey; the following is separate, qualitative evidence that the calibrated front end also operates correctly outside the bench. The calibrated front end has been exercised in field surface-wave surveys. Fig. 5 shows a multichannel waterfall across a 10–50 m line, with first-arrival moveout resolved at every offset using the higher PGA settings needed far-offset. This is field evidence of correct operation, not a controlled calibrated-vs-uncalibrated comparison, which remains future work.

## V. DISCUSSION

The method trades component precision for system-level measurement and digital assistance, unlike precision front ends characterized near the sensor’s Johnson-noise limit [15]. Its benefit is not that the ADC output has zero mean – a software subtraction could do that – but that the DACs modify the physical operating points before large downstream gains apply, so the correction is agnostic to the error source, provided the stage is measurable and the DAC has sufficient authority and range.

The foreground method was adopted after a continuously active DC servo showed a low corner centered impractically slowly, a faster loop interfered with the signal, and a gain change forced another transient – the corner-frequency/settling tradeoff reported for DC-servo offset compensation [16]. The proposed controller instead disconnects, holding DAC outputs constant throughout acquisition, adding no time-varying feedback or servo poles; EEPROM avoids the full transient after most gain changes. The method corrects the offset present when calibration runs, not drift directly; recalibrating at boot, gain change, or on command (Section III) limits drift accumulation. Remaining limitations are finite DAC resolution, no correction of noise or distortion, and unvalidated behavior under sustained vibration or thermal drift during convergence.

The controller runs as a pure integrator ( $k_{p,i} = 0$  in Eq. 3), not a shortfall: response speed is not a priority for a calibration that triggers once after deployment, so a one-time delay is acceptable. A full four-stage run completes in approximately 10 s typical (Fig. 4); the per-stage timeouts of Table I sum to 57.6 s worst case, reached only if every stage times out, which was not observed in the runs reported here.

Two independently assembled boards were characterized across the full gain range for this work; the headroom result of Table II holds on both despite markedly different default operating points, evidence the method is not tuned to one particular assembly. Repeated runs on both boards (Section IV) begin to separate algorithmic repeatability from board-to-board variation; a larger population and cross-temperature trials would be needed to generalize further. As noted in Section I, the comparison with prior self-calibration work [9], [11] is architectural, not quantitative, since those methods target the ADC transfer characteristic rather than each stage’s physical operating point.

## VI. CONCLUSION

A per-stage foreground self-calibration method was developed for a low-cost PSoC-based geophone front end. Four internal VDACS, an analog multiplexer, the acquisition delta-sigma ADC, an otherwise-idle hardware filter, and a configurable integral controller adjust each observable stage's operating point within the available DAC range and resolution. Unlike a SUM-only patch, the method addresses offsets before and after the summing node; unlike the continuously active servo tested during development, it adds no servo poles or settling-time-versus-distortion compromise. Stored gain-dependent corrections and automatic verification make full recalibration infrequent.

Oscilloscope measurements on two boards, across the PGA's full gain range ( $\times 1$  to  $\times 50$ ), show the front end consistently kept within 4% of the ADC's full-scale range after calibration, down from as much as 46% uncalibrated – the headroom result of Table II. The PGA stage alone shows only partial convergence under some conditions, traced to the loop's deadband gating its response at that stage specifically rather than to a control-loop or hardware limitation; this does not affect the overall headroom result, which is set by the stages closer to the ADC. Calibration converges in approximately 10 s typical (57.6 s worst case) per run, acceptable since triggered once after deployment; field surface-wave records over a 10–50 m line demonstrate operation, not a quantitative calibrated-vs-uncalibrated comparison. A larger board population, cross-temperature characterization, whether the injected VDAC references affect the noise floor, and a controlled field calibrated-vs-uncalibrated comparison remain future work.

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